

HYDROFARMS AGRI SOLUTIONS

EXCAVATION, DEWATERING, BACKFILLING AND COMPACTION

Activity-Specific Construction Method Statement

Document field	Controlled value
Document code	HF-GH01-MST-CIV-001
Revision	00 - Draft for coordination
Project	One-Feddan NFT Lettuce Greenhouse Project
Location	Alexandria, Egypt
Prepared by	Hydrofarms Agri Solutions
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Parent documents	HF-GH01-MST-GEN-001 and HF-GH01-MST-GEN-002

CONTROLLED EARTHWORK PROCEDURE
Excavation shall not start until the survey hold points are released. Formation shall not be covered until inspected. Buried work shall not be backfilled until testing, as-built surveying and concealed-work acceptance are complete.

Document Control

Approval and responsibility

Role	Primary responsibility
Project manager	Resources, programme, subcontractor control and approval status.
Construction manager	Work-front sequence, plant coordination and daily excavation release.
Civil engineer	Technical execution, formation assessment, fill selection and interfaces.
Geotechnical engineer	Soil interpretation, stability, suitability and corrective recommendations.
Surveyor	Excavation limits, formation levels, pipe inverts and as-built coordinates.
QA/QC engineer	ITP, material tests, density tests, inspection requests and records.
HSE manager	Excavation permit, access, shoring/slope, plant segregation and emergency controls.
MEP engineer	Sleeves, trenches, pipe bedding, testing and concealed-work release.
Client/consultant	Hold/witness point inspection and technical acceptance.

Revision history

Rev.	Date	Purpose	Description
00	15 Aug 2026	Draft for coordination	Initial project-specific English issue

1. Purpose

This method defines the controlled execution of stripping, excavation, temporary support, water control, formation treatment, selected filling, trench surround, backfilling and compaction for the production greenhouse, western service greenhouse and southern utility/tank zone.

The method protects the approved geometry, buried networks and foundation bearing conditions while preventing uncontrolled excavation, unsafe side collapse, contaminated fill and poorly compacted zones beneath floors or equipment.

2. Scope of Work

- Clearing, stripping and controlled site grading within the approved disturbance limits.
- Excavation for greenhouse foundations, pedestals and independent fan/cooling-pad frames.
- Excavation for production, reserve, nursery, germination and cooling-pad tank chambers where shown on AFC drawings.
- Excavation for the WTP base, equipment bases, sterilisation room and external utility structures where applicable.
- Trenches for 3-inch feed mains, 6-inch returns, WTP lines, pad-water lines, drainage, sleeves and earthing interfaces.
- Temporary dewatering, surface-water diversion and formation protection.
- Selected fill, bedding, pipe surround, trench backfill and structural backfill around underground rooms.
- Layer compaction, field density testing, level control and final earthwork records.

3. Governing Information

Priority	Controlled information
1	Approved geotechnical investigation and written formation recommendations
2	AFC civil, foundation, tank-room, drainage and external-works drawings
3	AFC coordinated buried-services routes, profiles, inverts and sleeve drawings
4	Approved survey control and setting-out releases under HF-GH01-MST-GEN-002
5	Approved material submittals, laboratory tests, ITP and risk assessment
6	Pipe/tank/equipment supplier requirements where stricter than this method

GEOTECHNICAL CONTROL

This method does not assign a universal safe excavation slope or bearing capacity. Side slopes, shoring and formation treatment shall follow the actual soil profile, groundwater condition, excavation depth and the approved geotechnical/temporary-works assessment.

4. Project Earthwork Zones

Zone	Typical work	Critical interface
Production greenhouse footprint	Local foundation excavation, floor preparation and buried distribution networks	Fifteen principal frame lines; table aisles and floor drainage
Western service greenhouse	Foundations, full floor build-up, germination/packaging utilities	Shared wall, central access and hygiene route
Southern production-tank area	Underground chambers for two operating production tanks	3-inch feed, 6-inch return, chillers and pump-removal access
Reserve-water room	Single linear room housing four adjacent 10 m ³ tanks	WTP inlet and transfer lines to operating tanks
Nursery tank area	Independent underground room for the confirmed 3 m ³ nursery tank	Nursery feed/return and 7.5 TR chiller interface
Cooling-pad systems	Underground chambers containing plastic tanks and pad-water pumps	Pad trough return, anti-virus screen footprint and maintenance access
WTP and external equipment	Equipment bases, interconnecting pipes and drainage	Incoming well line, reserve tanks and electrical/service access
Buried service corridors	Feed, return, drainage, water-treatment and auxiliary lines	Crossings, pipe inverts, sleeves and concealed-work survey

5. Resources

Personnel

- Civil engineer and competent excavation supervisor.
- HSE officer and trained banksman/plant marshal.
- Surveyor and assistant.
- Excavator/backhoe, loader and compaction-equipment operators with valid competency evidence.
- General workers trained in trench access, hand compaction and service identification.
- QA/QC technician and approved testing laboratory.

Plant and tools

Plant / tool	Selection requirement	Main use
Excavator/backhoe	Size compatible with access, excavation depth and nearby structures	Bulk and trench excavation
Loader / trucks	Covered/clean bodies as required; controlled routes	Haul, stockpile and disposal
Plate compactor / rammer	Suitable for confined areas and pipe surround	Trenches and room perimeters
Smooth-drum roller	Weight/vibration suitable for fill and substrate	Open-area layer compaction
Water bowser	Controlled spray and clean water	Moisture conditioning and dust control
Dewatering pumps	Duty/standby capacity, screened suction and safe discharge	Sump or surface-water removal
Shoring system	Engineered trench box, sheets/struts or proprietary system as assessed	Excavation-side support
Survey equipment	Calibrated total station and level	Limits, formation, layers and as-built
Density equipment	Approved calibrated method / laboratory	Field compaction verification

6. Material Requirements

Material	Minimum acceptance requirement	Intended use
Existing excavated soil	Used only after visual segregation and approved laboratory classification confirms suitability	General fill where approved
Selected structural fill	Well-graded, durable, free of organics/debris, controlled plasticity and approved by geotechnical engineer	Below floors, bases and structural zones
Pipe bedding/surround	Clean selected granular material compatible with pipe manufacturer; no damaging oversized particles	Below and around buried pipes
Granular sub-base	Approved grading, strength and compaction characteristics	Interlock/floor build-up and external service areas
Lean concrete/blinding	Approved mix and thickness to civil drawings	Clean formation and foundation working surface
Geotextile	Approved separation/filter grade where geotechnically required	Weak/wet formation or material separation

Prohibited fill

- Topsoil, vegetation, roots, rubbish, gypsum-rich/contaminated soil, construction waste or frozen material.
- Highly organic, expansive, collapsible or excessively wet material unless specifically treated and approved.
- Material containing particles incompatible with the lift thickness or capable of damaging pipes, tanks, waterproofing or membranes.

7. Preconditions and Permits

- HP-SUR-02 and relevant project setting-out releases are signed.
- Excavation permit identifies limits, depth, nearby services, side-stability method, access and emergency arrangements.
- Utility scan, drawing review and trial pits are complete where existing services may be present.
- The geotechnical report and formation acceptance process are available to the field team.
- Approved disposal and stockpile locations, haul routes and wheel-cleaning controls are established.
- Temporary drainage/dewatering discharge route is approved and will not erode soil, flood neighbours or contaminate drainage.
- Shoring, trench box or slope/batter requirements are determined before excavation reaches a hazardous depth.

8. Detailed Execution Method

8.1 Work-front preparation

- Brief the crew on the AFC drawing, method statement, risk assessment, permits, exclusion zones and hold points.
- Verify survey stakes and offset marks; fence the work and establish plant/pedestrian routes.
- Remove loose material, obstructions and vegetation only within approved limits.
- Provide surface-water diversion so runoff cannot enter open excavations.
- Photograph existing conditions and adjacent structures/services before work.

8.2 Clearing and stripping

- Strip topsoil/organic layer separately from reusable mineral soil; do not mix with structural fill stockpiles.
- Stockpile approved reusable material on stable ground with drainage and identification; keep away from excavation edges.
- Transport unsuitable or surplus material to the approved disposal location with documented quantities.
- Survey the stripped level before bulk grading or structural fill placement.

8.3 Excavation for foundations and structures

- Excavate in controlled stages to the approved plan limits and stop above final formation for careful trimming.
- Do not undercut adjacent foundations, control monuments, roads, buried services or property boundaries.
- Maintain a safe setback for spoil, plant and materials from excavation edges based on the stability assessment.
- Trim formation without disturbing bearing soil. Remove loose pockets and protect the surface immediately after acceptance.

- Where over-excavation occurs, do not replace with loose soil. Obtain an approved engineered repair using selected compacted fill or lean concrete as directed.

8.4 Tank-room and deep excavation

- Confirm the full chamber footprint, working space, waterproofing access, safe egress and future maintenance opening before excavation.
- Use battered sides or an engineered support system selected from the actual depth, soil and groundwater condition.
- Provide ladder/stair access, edge protection and inspection after rain, vibration, dewatering change or any sign of movement.
- Excavate and support in stages; install temporary works in accordance with the supplier/engineered sequence.
- Maintain the excavation dry enough to prevent base softening, piping or instability; do not pump in a manner that removes supporting fines.
- After formation acceptance, place blinding/base construction promptly and keep waterproofing interfaces clean.

8.5 Trench excavation

- Set centreline, width, crossing and invert from the coordinated route drawing; avoid unnecessary trench width.
- Use trial pits/hand digging near known or suspected services and protect exposed services from unsupported loading.
- Keep excavated material and plant outside the assessed edge zone; provide barriers and safe crossings.
- Trim trench bottom to design invert. Remove stones, hard spots and loose material that could create point loads.
- Provide approved bedding, place the pipe without forcing alignment and maintain access for jointing/testing.

8.6 Dewatering and water control

The civil/geotechnical engineer shall select the least disruptive system that maintains stable formation. The system may comprise surface diversion, local sump pumping, wellpoints or another engineered method depending on inflow and soil permeability.

- Install screened sumps away from bearing formation and pipe bedding where possible.
- Provide duty and standby pumps, non-return arrangements, secure hoses, power protection and high-water monitoring for critical excavations.
- Discharge through sediment control to an approved location; prevent erosion, flooding and recirculation into the excavation.
- Lower water gradually and monitor side/ground movement. Stop if boiling, piping, loss of fines or adjacent settlement appears.
- Continue dewatering until the permanent structure, waterproofing and backfill have sufficient stability to resist groundwater effects.

8.7 Formation inspection and treatment

- Survey formation level and invite geotechnical/QA-QC inspection before blinding or fill.
- Check for soft, disturbed, wet, organic or variable zones using the approved investigation/field method.
- Remove unsuitable pockets to the instructed extent and replace with approved compacted material or lean concrete.
- Use separation geotextile or stabilisation only where approved; record location, level and treatment.
- Protect accepted formation from traffic, rain and drying disturbance; re-inspect if conditions change.

8.8 Backfilling around tank rooms and structures

- Confirm concrete strength, waterproofing acceptance, protection layer and pipe/sleeve testing before backfill.
- Remove debris and standing water; place approved fill uniformly on opposing sides to avoid unbalanced wall loading.
- Use small compaction equipment adjacent to walls, waterproofing, tanks and pipes; maintain the approved equipment clearance.
- Place in controlled loose lifts not exceeding 200 mm unless a tested procedure demonstrates another approved thickness.
- Monitor wall/tank movement and stop if distortion, damage or abnormal settlement is observed.

8.9 Pipe bedding, surround and trench backfill

- Place a minimum 150 mm compacted bedding layer or the stricter pipe-supplier/detail requirement.
- Shape bedding to provide continuous support and keep joints accessible until inspection/testing is complete.
- Place selected surround evenly by hand on both sides, compact under haunches and continue to at least 300 mm above pipe crown.
- Do not use mechanical equipment that can displace or damage the pipe. Maintain at least the manufacturer-required cover before heavy compaction plant crosses.

- Install marker/identification provisions and complete the as-built survey before general backfill closes the route.
- Restore paved/interlock areas using the approved sub-base and surface build-up without creating future settlement strips.

8.10 Fill moisture conditioning and compaction

- Spread each layer uniformly and remove oversized or unsuitable inclusions.
- Condition moisture near the laboratory optimum range suitable for achieving density without pumping or rutting.
- Compact using a defined number of passes established by trial/initial testing for the material and equipment.
- Do not place the next lift until level, visual condition and required density tests for the current lift are accepted.
- Scarify and recondition failed areas, then retest; do not conceal a failed test by additional fill.

8.11 Final grading and protection

- Trim to approved subgrade/sub-base levels and maintain drainage falls away from structures and into approved drains.
- Protect accepted layers from contamination, excessive drying, rain, traffic rutting and later trench excavation.
- Survey finished levels and issue the earthwork completion report before floor/interlock works commence.

9. Compaction and Testing Criteria

The following criteria apply unless the approved geotechnical report or civil specification requires a higher standard. Laboratory maximum dry density and optimum moisture content shall be established for each materially different fill source.

Location	Acceptance criterion	Control basis
Structural fill below foundations/equipment bases	Minimum 95% of laboratory maximum dry density using the approved modified compaction reference	Each lift and each distinct work area
Below greenhouse floors/interlock/service areas	Minimum 95% of approved maximum dry density	Each lift; representative grid
Tank-room perimeter structural backfill	Minimum 95% with symmetric placement and no waterproofing damage	Each lift around each chamber
Trench backfill beneath paved/structural areas	Minimum 95%; pipe zone subject to pipe-manufacturer method	Each lift along route
Trench backfill in unloaded non-paved areas	Minimum 90% unless future use requires 95%	Each lift along route
Pipe bedding and surround	Firm, uniform support without pipe displacement; density per approved material/pipe procedure	Continuous visual plus selected tests
Moisture content	Within the range demonstrated to achieve density and stability	At density tests and when material condition changes

Minimum field-testing frequency

- Open-area fill: at least one field-density test per 250 m² per lift, with not fewer than three tests per major work area per lift.
- Trenches: at least one field-density test per 30 linear metres per lift and additional tests at chambers, crossings and reinstatement interfaces.
- Tank rooms and confined structural backfill: at least one test per side per lift or a more intensive consultant-approved pattern.
- Repeat tests at every visibly variable, repaired or failed area and whenever material source, moisture or compaction equipment changes.

FAILED TEST CONTROL

A failed density, moisture, level or formation result is a nonconformance for the affected area. Define its boundaries, rework the full affected lift, retest and obtain acceptance before adding another layer.

10. Inspection and Test Plan

No.	Inspection stage	Method	Type	Responsibility	Record
01	Approved drawings, survey release and excavation permit	Document review	H	Civil/HSE/QA-QC	Work-front release
02	Utility scan and trial pits	Field verification	H	Civil/MEP/HSE	Utility clearance
03	Fill source and laboratory properties	Laboratory testing	H	QA-QC/Consultant	Material approval
04	Stripped level / excavation limits	Survey and visual	W	Surveyor/Civil QA-QC	Survey sheet
05	Excavation side support / access	HSE inspection	H	HSE/Competent person	Excavation inspection
06	Formation level and condition	Survey/geotechnical inspection	H	Geotechnical/Consultant	Formation release
07	Dewatering and discharge	Functional/environmental check	S	Civil/HSE	Daily log
08	Pipe bedding and installed services	Visual/survey/test record	H	MEP/QA-QC/Consultant	Concealed-work release
09	Waterproofing/protection before backfill	Inspection	H	Civil/QA-QC/Consultant	Backfill release
10	Lift thickness, moisture and density	Survey/field test	W	QA-QC/Laboratory	Compaction report
11	Final subgrade/sub-base level	Survey and visual	H	Civil/Survey/Consultant	Earthwork completion
12	As-built and test dossier	Record review	R	QA-QC/Client	Turnover index

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

11. Hold Points

- HP-EW-01: Survey release, utility clearance and excavation permit before plant starts.
- HP-EW-02: Approval of any required slope/shoring/dewatering arrangement before hazardous-depth excavation.
- HP-EW-03: Geotechnical acceptance of foundation/tank-room formation before blinding or fill.
- HP-EW-04: Approval of buried services, tests and as-built survey before surround/backfill.
- HP-EW-05: Approval of waterproofing/protection and structural readiness before tank-room backfill.
- HP-EW-06: Acceptance of each failed/reworked compaction zone before the next lift.
- HP-EW-07: Final level and earthwork dossier acceptance before floors, interlock or foundations proceed.

12. HSE and Environmental Controls

Hazard	Consequence	Mandatory control	Residual risk
Side collapse	Burial/crushing	Engineered batter/shoring, competent-person inspections, no entry after change until rechecked	Medium
Falls into excavation	Serious injury	Barriers, edge protection, lighting, signed access and safe crossing	Medium
Plant movement	Collision/crushing	Segregated routes, banksman, alarms/cameras and exclusion zone	Medium
Unknown services	Electrocution/flood/release	Permit, scan, trial pits, hand dig and service isolation/protection	Medium
Water ingress	Instability/drowning/equipment loss	Diversion, duty/standby pumping, monitoring and evacuation trigger	Medium
Confined/deep chamber	Restricted escape / atmosphere	Safe access; confined-space assessment, permit and rescue plan where criteria apply	Medium
Spoil/edge surcharge	Collapse	Approved setback and stockpile height; no plant parked at edge	Low
Dust/heat	Respiratory/heat stress	Water suppression, suitable PPE, hydration and work/rest regime	Low
Compaction vibration	Structure/service damage	Equipment selection, clearance and monitoring	Low

Daily excavation inspection

- At shift start and after rain, dewatering interruption, vibration, support change or visible movement.
- Check sides/supports, access, barriers, spoil setback, water, adjacent services and atmospheric requirements.

- Record the inspection. Remove personnel and plant immediately if stability is uncertain.

Environmental controls

- Use approved disposal areas and prevent mud tracking to public roads.
- Control dust without saturating formation or fill.
- Treat/sediment dewatering discharge as required; no uncontrolled nutrient, oil or silt discharge.
- Prevent fuel/oil storage within drainage paths or excavation-edge zones; provide spill response.
- Minimise open-excavation duration and restore disturbed ground promptly.

13. Records and Handover

Record	Minimum content	Release stage
Excavation permit	Location, depth, services, support, access, dewatering and signatures	Before excavation
Daily excavation inspection	Sides/supports, water, access, plant/spoil setback and weather	Each shift / after change
Material approval	Source, classification, grading, MDD/OMC and suitability	Before fill delivery/use
Formation inspection	Coordinates, level, soil condition, treatment and acceptance	Before blinding/fill
Dewatering log	Pump status, water level, discharge condition and incidents	During dewatering
Concealed-work release	Pipe/sleeve test, photographs, route/invert and as-built file	Before backfill
Compaction report	Area/lift, material, moisture, density, result and retest	Before next lift
Final earthwork survey	Subgrade/sub-base levels, falls and deviations	Before subsequent works
Earthwork dossier	All approvals, tests, surveys, NCR closure and disposal records	Package handover

Appendix A - Excavation Daily Checklist

- Current excavation permit and AFC drawing available.
- Survey marks/offsets intact and limits clearly identified.
- Known services located, exposed/protected and marked.
- Side slope/shoring matches the approved arrangement.
- Safe access, barriers, lighting and emergency route are maintained.
- Spoil, materials and plant remain outside the approved edge setback.
- Water level/dewatering system is controlled and discharge is acceptable.
- No cracking, sloughing, bulging, boiling, piping or adjacent settlement observed.
- Weather and work changes assessed before entry continues.

Appendix B - Compaction Lot Register Template

Lot ID	Area / chainage	Lift level	Material source	Test result	Status / retest
EW-LOT-001	Recorded work location	Recorded lift / RL	Approved source code	Density and moisture	Accepted / reworked
EW-LOT-002	Recorded work location	Recorded lift / RL	Approved source code	Density and moisture	Accepted / reworked
NEXT METHOD STATEMENTS					
The next coordinated civil submissions are HF-GH01-MST-CIV-003 for underground tank rooms/waterproofing and HF-GH01-MST-CIV-004 for buried utilities. Their approved details shall be available before the associated excavations are released.					